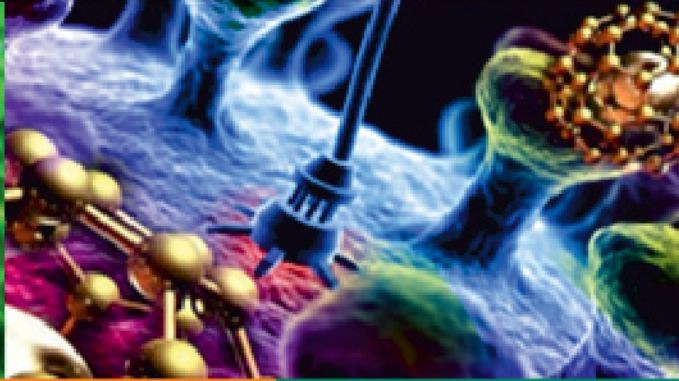
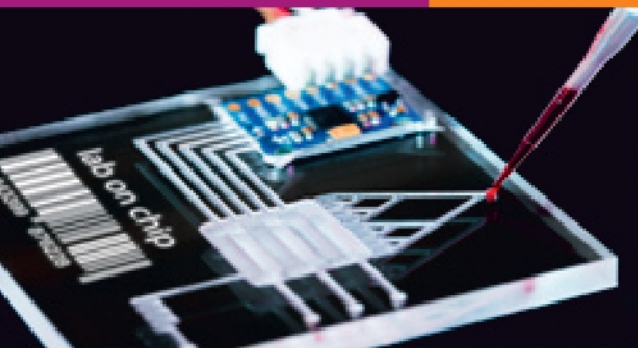


MIT ART, DESIGN AND TECHNOLOGY UNIVERSITY
MIT School of Bioengineering Sciences & Research

7th INTERNATIONAL CONFERENCE
ON
RECENT TRENDS IN BIOENGINEERING
(ICRTB 2024)

January 19-20, 2024

ABSTRACTS



**MIT School of
Bioengineering Sciences
& Research**
Breathe Life into Engineering Career!



**MIT-ADT
UNIVERSITY**
PUNE, INDIA
A Leap Towards The World Class Education

PP-014

Aptamer tethered hetero-polymer conjugated mesoporous silica nanoparticles (MSNPs) mediated delivery of dual siRNA to induce apoptosis in breast cancer cells

Niladri Haldar and Virendra Gajbhiye

Nano bioscience, Agharkar Research Institute, Pune-411038, India

Abstract:The study aimed to explore the gene silencing potential of multifunctional aptamer-tethered hetero-polymer conjugated mesoporous silica nanoparticles (MSNPs) loaded with dual small interfering RNAs (siRNAs) in MCF-7 breast cancer cells. Objectives included synthesizing and characterizing hetero-polymer conjugated MSNPs, attaching polyethylene glycol (PEG) and MUC-1 aptamer onto the Nano carriers, and evaluating gene silencing efficacy. MSNPs were synthesized via sol-gel technique, conjugated with hetero-polymer molecules, and subsequently modified with PEG and MUC-1 aptamer. Characterization involved dynamic light scattering (DLS), scanning electron microscopy (SEM), and Fourier-transform infrared spectroscopy (FTIR). Quantitative real-time PCR assessed the gene silencing efficacy of dual siRNAs (MCL-1 and Survivin) loaded onto the aptamer-modified MSNPs in MCF-7 cells, complemented by a cell death assay. FTIR and zeta potential measurements confirmed successful hetero-polymer conjugation onto MSNPs. Targeted Nano carriers significantly suppressed the target genes compared to non-targeted ones and lipofectamine 2000. Cell death assays verified apoptosis induction through Caspase assays. In conclusion, aptamer-modified hetero-polymer conjugated MSNPs hold promise for delivering therapeutic agents in MCF-7 cells. Dual targeting of anti-apoptotic genes exhibited enhanced efficacy in inhibiting cancer growth compared to single-gene targeting, suggesting potential avenues for future targeted breast cancer therapies.

Keywords: Gene silencing, Aptamer, siRNA, Mesoporous silica nanoparticles, Targeted therapy, Dual siRNA, Breast cancer, MUC-1 aptamer.

MIT School of Bioengineering Sciences & Research

Bioengineering is a multidisciplinary field that has gained immense importance in the global scenario. MIT School of Bioengineering Sciences & Research is a constituent unit of MIT Art, Design and Technology University that aims to prepare students for ambitious careers in bioengineering by imparting education and hands-on training through project-based learning. The school helps in introducing students to the world of untapped opportunities in biology using engineering skills. The school envisions to empower India to become a leader in Bioengineering research, development and manufacturing sectors. It offers B. Tech. (Bioengineering), Int. M. Tech. (Bioengineering), M. Tech. (Environmental Bioengineering), M.Sc (Industrial Biotechnology) and Ph.D Programs in Biotechnology, Biomedical Engineering, Bioinformatics and Biomaterials. The super-specialized courses include Pharmaceutical Engineering, Synthetic Biology, Big Data Analytics, Biosensors and IoT, Biomedical Imaging, Biomechanics, Nano-Biotechnology, Robotics, Machine Learning and Artificial Intelligence, Environmental bioengineering, etc. The school is equipped with state-of-art laboratories for carrying out high-end research and innovative product development.



Rajbaug, Next to Hadapsar, Loni Kalbhor, Pune 412201, India.

+91 744 7444 027 / 28 | +91 914 6051 841 / 42

www.mitbio.edu.in / icrtb@mituniversity.edu.in

ISBN 978-93-6039-051-8

